This note goes through the detailed calculation of the quantum corrections to the mass of the kink, the $1+1$ dimensional soliton. It is inspired by (and closely follows) Chapter 2.2 of [1]. You will find more information in the source but I have filled in many details that the book has skipped over.

Consider the $1+1$ dimensional action

$$S = \int d^2x \left[\frac{1}{2}(\partial_\mu \phi)^2 - V(\phi)\right] \tag{1}$$

with the potential

$$\begin{aligned} V(\phi) &= -\frac{m}{2}\phi^2 + \frac{\lambda}{4}\phi^4 + \frac{\lambda}{4}v^4 \\ &= \frac{\lambda}{4}(\phi^2 - v^2)^2. \end{aligned} \tag{2}$$

For a static field $\partial_t \phi = 0$ the energy functional

$$E = \int dx \left((\partial_x \phi)^2 + V(\phi)\right) \tag{3}$$

is minimized for the kink solution

$$\phi_k(x) = v \tanh\left(\frac{m}{\sqrt{2}}(x - x_0)\right) \tag{4}$$

which satisfied the classical Euler-Lagrange equation. For now this is just a field configuration, however it does satisfy $V(\phi_k(\pm\infty)) = V_{\min} = 0$, which is why we say it is a vacuum configuration. The only other such configurations we know are the trivial zero vacuum $\phi(x) = 0$ (which is the vacuum we usually work with in elementary quantum field theory) and the trivial non-zero vacuum $\phi(x) = \text{const} \neq 0$. This new vacuum ϕ_k is non-trivial in the sense that has spatial dependence which carries with it important consequences.
Using eq. (4), we can compute the energy which evaluates to

$$E_{cl} = \frac{2\sqrt{2}}{3}\frac{m^3}{\lambda}. \tag{5}$$

The meaning of the subscript *cl* will become clear soon. This equation is already pretty interesting. Note that this is the energy of a solution which we obtained by minimizing the energy, which means that eq. (5) is the lowest possible energy a kink can have. We can Lorentz-boost eq. (4) which would give it additional energy coming from its momentum. Therefore, we can identify E_{cl} with its mass M_{cl} and write $E_{cl} = \sqrt{M_{cl}^2 + k^2}$ where k is the spatial momentum. In this sense, the kink acts particle-like, but this is misleading. The object stems from a field, and even though the field is in a localized particle-like configuration, the fact that $S \sim \frac{1}{\lambda}$ implies that the theory is already non-perturbative and will thus not contain any small particle-like excitations. In that sense we cannot treat the kink as a particle.
We can, however, study small excitations *around* the kink. As mentioned above, the kink cannot act as a particle but merely a non-trivial vacuum. So in the presence of a kink, excitations are quantized in the non-trivial background

$$\phi(t, x) = \phi_k(x) + \eta(t, x). \tag{6}$$

The energy of the kink given in eq. (5) is classical. When excitations like η are present around the kink background, they will scatter with the localized lump and change its energy, and hence its mass. Quantum mechanically, the vacuum has non-zero energy and the kink slowly

interpolates between pure vacua at spatial infinity. To define its full mass, we must therefore subtract the vacuum energy since the kink is defined above the pure vacuum,

$$M = E_K - E_V \tag{7}$$

where E_K is the full quantum mechanical energy of a kink at rest and E_V is the vacuum energy. Of course, classically $E_V = 0$ and $E_K = E_{cl}$ which gives $M = E_{cl} = M_{cl}$ as expected. Now let us see what components enter E_V and E_K. The vacuum energy is

$$E_V = E_V^{cl} + E_V^{ZP} + \delta E + E_V^{ct}.$$

E_V^{cl} is the classical energy of the vacuum which as already mentioned is zero. E_V^{ZP} is the energy of all the zero-point oscillators coming from

$$\sum_i \left(n_i + \frac{1}{2}\right)\omega_V^i$$

for $n_i = 0$. This is of course divergent which means that we must renormalize it by adding a counterterm E_V^{ct}. Finally, δE is the cosmological constant. For the kink the picture is similar:

$$E_K = E_K^{cl} + E_K^{ZP} + \delta E + E_K^{ct}.$$

The kink background has its own zero-point energy (since the space-dependent configuration modifies the zero-point), and therefore its own counterterms. δE is of course the same, the cosmological constant contributes equally to all energies. And it will obviously cancel in the difference as expected. Therefore, we have to only calculate three terms in

$$M = \Delta E_{cl} + \Delta E_{ZP} + \Delta E_{ct}. \tag{8}$$

The first one is the easiest,

$$\Delta E_{cl} = M_{cl}.$$

The third one requires a bit more work but it is still straight-forward. Notice that $\lambda\phi^4$ is superrenormalizable in $d = 1 + 1$ dimensions. In a perturbative theory around the trivial vacuum, the mass correction is

$$m^2 + i3\lambda \int \frac{d^2k}{(2\pi)^2}\frac{1}{k^2 - 2m^2} + \delta m^2$$

and picking

$$\delta m^2 = \frac{3\lambda}{2\pi}\int_0^\infty dk \frac{1}{\sqrt{k^2 + 2m^2}}$$

sets the entire divergent diagram to zero. In fact, the 2-point self-energy is the only divergent diagram in this theory.

Unfortunately, while we can use perturbative methods to find the counterterms, we cannot do the same to calculate ΔE_{ZP}, even though calculating the difference between the vacuum zero-point energy and the zero-point energy of an excitation and then renormalizing amounts to calculating the self-energy. If we attempt to insert eq. (6) into the full action and expand around η in a standard Feynman diagram way, we will obtain vertices that are space-dependent which will effectively kill momentum conservation making the Feynman rules significantly harder to work with. The difference to effective field theory, where we do essentially the same procedure but obtain vertices constant in space and time, is that there we have clear separation of scales and can assume that in the soft regime the hard modes do not oscillate and merely act as

constant backgrounds. Here, however, we are calculating the corrections to the kink mass, and if we assumed the background to be constant we would be back calculating the corrections to the mass of the excitation in the constant vacuum.
Hence, our final and most difficult task is competing ΔE_{ZP} non-perturbatively. To do this, we go back to the Lagrangian and insert eq. (6). But instead of expanding it in weak coupling, we solve it explicitly. This can be done on a discrete lattice. But we first we want to understand the intuition.
Take a one-dimensional space of length a. The wavefunction $\psi(x)$ of the small excitation η around the kink travels along this space. It starts at $\psi(0)$ and ends at $\psi(a)$. Eventually we will take $a \to \infty$ so at the finite a we don't want to have any non-trivial effects like bouncing off a potential wall or reflection. To avoid these we set the boundary condition $\psi(0) = \psi(a)$ which effectively identifies the two points.
At some point, the wavefunction will encounter this localized lump and scatter off of it. If we set $x_0 = 0$ in eq. (4), this point will be at the center but the exact location of the point does not matter as long as it's far away from a. Notice that due to energy conservation, the wavefunction cannot change its energy after the scattering off of the kink. The only thing that can change is its phase which we will denote by $\delta(k)$ where k is the wave vector. This means that the total phase change of the wave in the trivial vacuum is e^{ika} while in the kink vacuum it's $e^{i(ka+\delta(k))}$. The dispersion relations are simply given by

$$\omega_{K,V} = \sqrt{k_{K,V}^2 + \mu^2},$$

where $\mu := \sqrt{2}m$ is the Lagrangian mass of the excitation. When we discretize the space, the wave vectors also become discretized, such that the periodic boundary conditions now enforce

$$\begin{aligned} k_V^n a &= 2\pi n \\ k_K^n a + \delta(k_K^n) &= 2\pi n. \end{aligned} \tag{9}$$

Since the kink is localized far away from the boundary, at the boundary the phase shift of the kink wave vector will not be too different from that of the vacuum wave vector,

$$\delta(k_K^n) \approx \delta(k_V^n). \tag{10}$$

In fact, in general we might assume that the phase shift is infinitesimal in the limit $a \to \infty$. Discretization turns the continuous space of length a into $2N$ points separated by the distance $\frac{a}{2N}$. This also means that instead of infinitely many modes, there will only be $2N$ modes, each oscillating with the wavenumber k_n. ΔE_{ZP} becomes

$$\Delta E_{ZP} = \frac{1}{2} \sum_{n=-N}^{N} (\omega_K^n - \omega_V^n) \tag{11}$$

Let us imagine that the kink spectrum consists of two discrete states of eigenvalues ω_K^0 and ω_K^1 and a continuum (this will be shown later, the argument that follows works w.l.o.g.). When the wavefunction scatters off the kink in the discrete modes of oscillation, it itself will become a bound state. That is to say, its two lowest normal modes corresponding to ω_V^0 and ω_V^1 will shift lower to modes corresponding to ω_K^0 and ω_K^1. But in the vacuum $n = 0, 1$ were already the lowest modes so in the kink background these will now correspond to the modes $n = -1, 0$. Actually, in the vacuum $\omega_V^0 = 0$ is the zero mode, so the lowest states of the trivial vacuum are $n = -1, 1$. Thus decomposing the sum into discrete and continuum parts (where the continuum starts at $n = \pm 2$) and using the fact that the dispersion relations are symmetric in n we obtain

$$\Delta E_{ZP} = \frac{1}{2}(\omega_K^0 - \omega_V^{-1}) + \frac{1}{2}(\omega_K^1 - \omega_V^1) + \underbrace{\sum_{n=2}^{N} (\omega_K^n - \omega_V^n)}_{\Delta E_{ZP}^c}.$$

For the continuum part we plug in eqs. (9) and (10) to get

$$\begin{aligned}
\Delta E^c_{ZP} &= \sum_{n=2}^{N}\left(\sqrt{\mu^2+(k^n_K)^2}-\sqrt{\mu^2+(k^n_V)^2}\right)\\
&= \sum_{n=2}^{N}\left(\sqrt{\mu^2+(k^n_V)^2-\frac{2\delta k^n_V}{a}}-\omega^n_V\right)\\
&= -\underbrace{\frac{1}{a}\sum_{n=2}^{N}}_{\to\frac{1}{2\pi}\int_0^\infty dk}\frac{\delta k^n_V}{\omega^n_V}+\mathcal{O}(a^{-2})\\
&\to -\frac{1}{2\pi}\int_0^\infty dk\frac{k\delta(k)}{\sqrt{k^2+\mu^2}}.
\end{aligned} \tag{12}$$

The lowest modes of the vacuum have eigenvalues $\omega_V^{-1}=\omega_V^1=2\sqrt{m}$ so all that remains now (except for renormalization) is calculating $\delta(k)$ and the discrete eigenvalues of the kink.
To solve for the eigenvalues, let us write down the full Lagrangian of an excitation η around the kink background. Given the action $S[\phi]=S[\phi_k+\eta]$,

$$\begin{aligned}
S[\phi] &= \int d^2x\left[\frac{1}{2}(\partial_t\eta)^2-\frac{1}{2}(\partial_x\phi_k+\partial_x\eta)^2-V(\phi_k+\eta)\right]\\
&= \int d^2x\left[\frac{1}{2}(\partial_\mu\eta)^2-\frac{1}{2}V''(\phi_k)\eta^2\right]-\int d^2x\left[\frac{1}{2}(\partial_x\phi_k)^2+V(\phi_k)\right]+\mathcal{O}(\eta^3)
\end{aligned}$$

from which it is obvious that small oscillations of η are governed by the first term which is a quadratic Lagrangian. The corresponding equation of motion is

$$\partial_t^2\eta=-\partial_x^2\eta+V''(\phi_k)\eta. \tag{13}$$

In other words, the equation

$$\partial_t^2\eta(t,x)=O(x)\eta(t,x),$$

where

$$O(x)\coloneqq-\frac{d^2}{dx^2}+U(x)$$

and $U(x)\coloneqq V''(\phi_k(x))$ is linear and its normal modes oscillate independently which implies

$$\eta(t,x)=e^{-i\omega_n t}f_n(x). \tag{14}$$

Inserting this into eq. (13) we obtain the Schrödinger-type eigenvalue equation

$$\left(-\frac{d^2}{dx^2}+U(x)\right)f_n(x)=\omega_n^2 f_n(x). \tag{15}$$

Since from eq. (1) the equation of motion for ϕ_k is

$$\frac{d^2\phi_k}{dx^2}=\frac{dV}{d\phi_k},$$

we see that

$$\left(-\frac{d^2}{dx^2}+V''(\phi_k)\right)\frac{d\phi_k}{dx}=0,$$

which implies that there is an eigenfunction $f_0(x) = \frac{d\phi_k}{dx}$ with eigenvalue $\omega_0 = 0$. Since $f_0(x) \neq 0$, it has no nodes and by the node-theorem this must be the eigenfunction with the lowest eigenstate. This allows us to identify $\omega_0 = \omega_K^0 = 0$.
The second discrete eigenfunction and the continuum can be found analytically as well but we will skip these steps since these are rather technical and bear no insight. There is a second discrete state with eigenvalue $\omega_1 = \sqrt{\frac{3}{2}}m = \omega_K^1$. The continuum solutions are given by

$$f_k(x) = e^{ikx}\left[3m^2\tanh^2\left(\frac{m(x-x_0)}{\sqrt{2}}\right) - m^2 - 2k^2 - 3\sqrt{2}imk\tanh\left(\frac{m(x-x_0)}{\sqrt{2}}\right)\right]$$

which has the $x \to \pm\infty$ limit

$$f_k(x) = \sqrt{4(m^2-k^2)^2 + 18m^2k^2}\exp\left\{i\left(kx \mp \arctan\left(\frac{3\sqrt{2}}{2}\frac{mk}{m^2-k^2}\right)\right)\right\}.$$

We read off the phase shift $\delta(k)$,

$$\delta(k) = -2\arctan\left(\frac{3\sqrt{2}}{2}\frac{mk}{m^2-k^2}\right).$$

We can now solve eq. (12). Using integration by parts,

$$\begin{aligned}
\Delta E_{ZP}^c &= \frac{1}{2\pi}\int_0^\infty \delta(k)\frac{d}{dk}\left(\sqrt{k^2+\mu^2}\right) \\
&= \frac{1}{2\pi}\left[\delta(k)\sqrt{k^2+\mu^2}\right]_0^\infty - \frac{1}{2\pi}\int_0^\infty dk\sqrt{k^2+\mu^2}\frac{d\delta}{dk} \\
&= \underbrace{\frac{1}{2\pi}\left[-2\arctan\left(\frac{3\sqrt{2}}{2}\frac{mk}{m^2-k^2}\right)\sqrt{k^2+\mu^2}\right]_0^\infty}_{-\frac{3m}{\sqrt{2}\pi}+\sqrt{2}m} - \frac{1}{2\pi}\int_0^\infty dk\sqrt{k^2+\mu^2}\frac{d\delta}{dk} \\
&= -\frac{3m}{\sqrt{2}\pi} + \sqrt{2}m - \frac{1}{2\pi}\int_0^\infty dk\sqrt{k^2+2m^2}(-3\sqrt{2}m)\frac{m^2+k^2}{(k^2+2m^2)(k^2+\frac{1}{2}m^2)} \\
&= -\frac{3m}{\sqrt{2}\pi} + \sqrt{2}m + \frac{3\sqrt{2}}{2\pi}m\int_0^\infty dk\left[\frac{m^2}{\sqrt{k^2+2m^2}(k^2+\frac{1}{2}m^2)} + \frac{k^2}{\sqrt{k^2+2m^2}(k^2+\frac{1}{2}m^2)}\right] \\
&= -\frac{3m}{\sqrt{2}\pi} + \sqrt{2}m + \frac{3\sqrt{2}}{2\pi}m\int_0^\infty dk\left[\frac{\frac{1}{2}m^2}{\sqrt{k^2+2m^2}(k^2+\frac{1}{2}m^2)} + \frac{1}{\sqrt{k^2+2m^2}}\right] \\
&= -\frac{3m}{\sqrt{2}\pi} + \sqrt{2}m + \frac{3\sqrt{2}}{4\pi}m^3\underbrace{\int_0^\infty dk\frac{1}{\sqrt{k^2+2m^2}(k^2+\frac{1}{2}m^2)}}_{\frac{2\sqrt{3}}{9}\frac{\pi}{m^2}} + \frac{3\sqrt{2}}{2\pi}m\int_0^\infty \frac{1}{\sqrt{k^2+2m^2}} \\
&= \left(-\frac{3}{\sqrt{2}\pi} + \sqrt{2} + \frac{\sqrt{6}}{6}\right)m + \frac{3\sqrt{2}}{2\pi}m\int_0^\infty dk\frac{1}{\sqrt{k^2+2m^2}}.
\end{aligned} \tag{16}$$

For the counterterms we get

$$\begin{aligned}
\Delta E_{ct} &= \int dx\frac{1}{2}\delta m^2(\phi_k(x)^2 - v^2) \\
&= \frac{1}{2}\delta m^2\int_{-\infty}^\infty dx v^2\left(\tanh^2\left(\frac{m}{\sqrt{2}}x\right) - 1\right) \\
&= -\frac{\sqrt{2}m}{\lambda}\delta m^2 \\
&= -\frac{3\sqrt{2}}{2\pi}m\int_0^\infty dk\frac{1}{\sqrt{k^2+2m^2}},
\end{aligned}$$

which exactly cancels the divergent part in eq. (16). Putting together the discrete eigenvalues, we finally obtain

$$\begin{aligned} M &= \frac{2\sqrt{2}}{3}\frac{m^3}{\lambda} + \left(\frac{\sqrt{6}}{4} - \sqrt{2}\right) m + \left(-\frac{3m}{\sqrt{2}\pi} + \sqrt{2}m + \frac{\sqrt{6}}{6}\right) \\ &= \frac{2\sqrt{2}}{3}\frac{m^3}{\lambda} + \left(\frac{\sqrt{6}}{12} - \frac{3\sqrt{2}}{2\pi}\right) m \end{aligned}$$

which is the mass correction up to terms coming from $\mathcal{L}_{\text{int}}$, i.e. terms of order λ.

References

[1] E. J. Weinberg, *Classical solutions in quantum field theory: Solitons and Instantons in High Energy Physics*, Cambridge Monographs on Mathematical Physics (Cambridge University Press, Sept. 2012).